

London Labour Market — Data Analyst Portfolio Case Study

Professional BI / Data Analyst Portfolio | End-to-end labour-market decision-support project

1. Executive Summary

An evidence-based decision-support framework for understanding variation in labour-market conditions across London boroughs, integrating seven datasets with data-quality assessment, segmentation, priority scoring, sensitivity testing, intervention mapping and Power BI.

2. Business Problem

London boroughs experience labour-market conditions differently. The challenge is to identify priority areas transparently while retaining context, uncertainty and interpretability.

3. Analytical Questions

Identify relative labour-market pressure; segment borough profiles; test robustness; map profiles to intervention themes; communicate results through interactive decision support.

4. Data & Preparation

Seven source datasets were integrated. Borough identifiers were standardised, completeness and duplicates assessed, and validated borough-level outputs created. The master geography retains 33 borough records; the main analytical population contains 32 observations where required indicators are available.

5. Analytical Workflow

Source inspection → data quality → joins → borough analytical dataset → exploratory analysis → feature standardisation → clustering/validation → priority scoring → sensitivity/robustness → intervention mapping → Power BI → QA/reconciliation → stakeholder testing.

6. Segmentation & Clustering

Clustering is descriptive segmentation, not a causal model. Cluster profile and priority score are deliberately kept separate.

7. Priority Framework

A transparent relative priority measure identifies labour-market pressure while remaining explainable to non-technical stakeholders.

8. Robustness & Sensitivity

Alternative weighting scenarios distinguish stable findings from assumption-sensitive rankings and reduce false precision.

9. Intervention Mapping

Profiles are linked to evidence-informed intervention themes; they are decision-support prompts, not guaranteed policy effects.

10. Power BI Decision-Support Product

Three views: Executive Overview, Borough Diagnostic, and Priority & Intervention. The model separates dimensions and analytical outputs; DAX supports KPIs, ranking, tiers and robustness.

11. Governance & QA

Methodology, KPI definitions, reproducibility, data quality, reconciliation, accessibility and stakeholder usability are documented. Validated analytical outputs are the reference layer.

12. Key Analytical Judgement

The project prioritises transparent assumptions, separation of concepts, sensitivity testing, causal caution and uncertainty communication over unnecessary black-box complexity.

13. Business Value

Repeatable prioritisation; explicit assumptions; segmentation context; robust-vs-sensitive findings; interactive decision support; auditable evidence trail.

14. Limitations

Observational analysis; thorough aggregation; dependence on indicators/scaling/weights; clustering sensitivity; missing observations; need for policy and qualitative evidence.

15. Skills Demonstrated

SQL, Python, statistics, clustering, sensitivity analysis, KPI design, Power BI, DAX, UX, QA, governance and stakeholder communication.

16. Interview Story

Seven datasets → validated thorough model → SQL/Python analysis → descriptive clustering → transparent priority framework → sensitivity testing → intervention mapping → Power BI decision support.

17. Recommended Portfolio Evidence

Use dashboard screenshots, Python and SQL excerpts, methodology/robustness visuals and QA evidence.

18. Conclusion

The project demonstrates an end-to-end analytical engagement in which technology accelerates delivery while human judgement remains responsible for interpretation, validation and recommendations.